

UEC754: IMAGE PROCESSING AND COMPUTER VISION

L	T	P	Cr
2	0	2	3.0

Course Objective:

To make students understand image fundamentals and how digital images can be processed, Image enhancement techniques and its application, Image compression and its applicability, fundamentals of computer vision, geometrical features of images, object recognition and application of real time image processing.

Introduction: Digital image representation, fundamental steps in image processing, elements of digital image processing systems digitization.

Digital Image fundamentals: A Simple Image Model, Sampling and Quantization, Relationship between Pixel, Image Formats, Image Transforms.

Image Enhancement: Histogram processing, image subtraction, image averaging, smoothing filters, sharpening filters, enhancement in frequency and spatial domain, low pass filtering, high pass filtering.

Image Compression: Fundamentals, Image Compression Models, Elements of Information Theory, Error-Free Compression, Lossy Compression, Recent Image Compression Standards.

Computer Vision: Imaging Geometry; Coordinate transformation and geometric warping for image registration, Hough transforms and other simple object recognition methods, Shape correspondence and shape matching, Principal Component Analysis, Shape priors for recognition.

Familiarization with standards: IEEE 610.4-1990. IEEE 2020-2024 (Automotive Imaging), IEEE 1858-2023 (Camera Phone Image Quality), JPEG 2000 (ISO/IEC 15444), JPEG XR (ISO/IEC 29199-2), JPEG AI.

Laboratory Work:

1. Introduction to image processing on MATLAB.
2. Image effects based on image quantization.
3. Image enhancement algorithms for histogram processing, filtering.
4. Fourier transform of images and filtering in frequency domain.
5. Realization of any one image compression algorithm.
6. Introduction to computer vision tools.

Minor Project: Image Compression and Facial Feature Detection with FPGA/ASIC/ARM/DSP Processors.

Course Learning Objectives (CLO): The students will be able to:

1. Illustrate the fundamentals of image processing.
2. Apply image enhancement techniques in spatial and frequency domain.
3. Analyze the mathematical modeling of image restoration and compression.
4. Demonstrate real-time image processing with computer vision.

Text Books

1. Gonzalez, R.C., and Woods, R.E., Digital Image Processing, Dorling Kingsley (2009) 3rd ed.
2. Jain A.K., Fundamentals of Digital Image Processing, Prentice Hall (2007).
3. Sonka M., Image Processing and Machine Vision, Prentice Hall (2007) 3rd ed.
4. D. Forsyth and J. Ponce, Computer Vision - A modern approach, Prentice Hall.

5. B. K. P. Horn, Robot Vision, McGraw-Hill.
6. E. Trucco and A. Verri, Introductory Techniques for 3D Computer Vision, Prentice Hall.
7. Richard Szeliski, Computer Vision: Algos and Applications, Springer.

Reference Books

1. Tekalp A.M., Digital Video Processing, Prentice Hall (1995).
2. 2. Ghanbari M., Standard Codecs: Image Compression to Advanced Video Coding, IET Press (2003).